

BPM 1

FTM: Lecture 39

Autonomic Integration ***Flipped Classroom***

Subramanya Upadhyaya, MD.

Professor

Department of Physiology, Neuroscience & Behavioral Sciences

School of Medicine, St. George's University

Email: supadhyaya@sgu.edu

Copyright

All year 1 courses materials, whether in print or online, are protected by copyright. The work, or parts of it, may not be copied, distributed or published in any form, printed, electronic or otherwise.

As an exception, students enrolled in year 1 of St. George's University School of Medicine and their faculty are permitted to make electronic or print copies of all downloadable files for personal and classroom use only, provided that no alterations to the documents are made and that the copyright statement is maintained in all copies.

View only files, such as lecture recordings, are explicitly excluded from download and creating copies of these recordings by students and other users are strictly illegal.

The author of this document has made the best effort to observe current copyright law and the copyright policy of St George's University. Users of this document identifying potential violations of these regulations are asked to bring their concern to the attention of the author.

REQUIRED READING MATERIAL BEFORE LECTURE 39

❖ You are required to watch the video before attending L39 with the below link:

<https://sgu1.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=c99731bf-c699-41e4-8264-b34e00d4756a>

❖ Read related material in the recommended Physiology textbook, chapter 6-Autonomic Nervous System, Under heading, “AUTONOMIC INTEGRATION”. Use link below to reach here:

<https://premiumbasicsciences-lwwhealthlibrary-com.sgu.idm.oclc.org/content.aspx?sectionid=253086161&bookid=32>

LEARNING OBJECTIVES

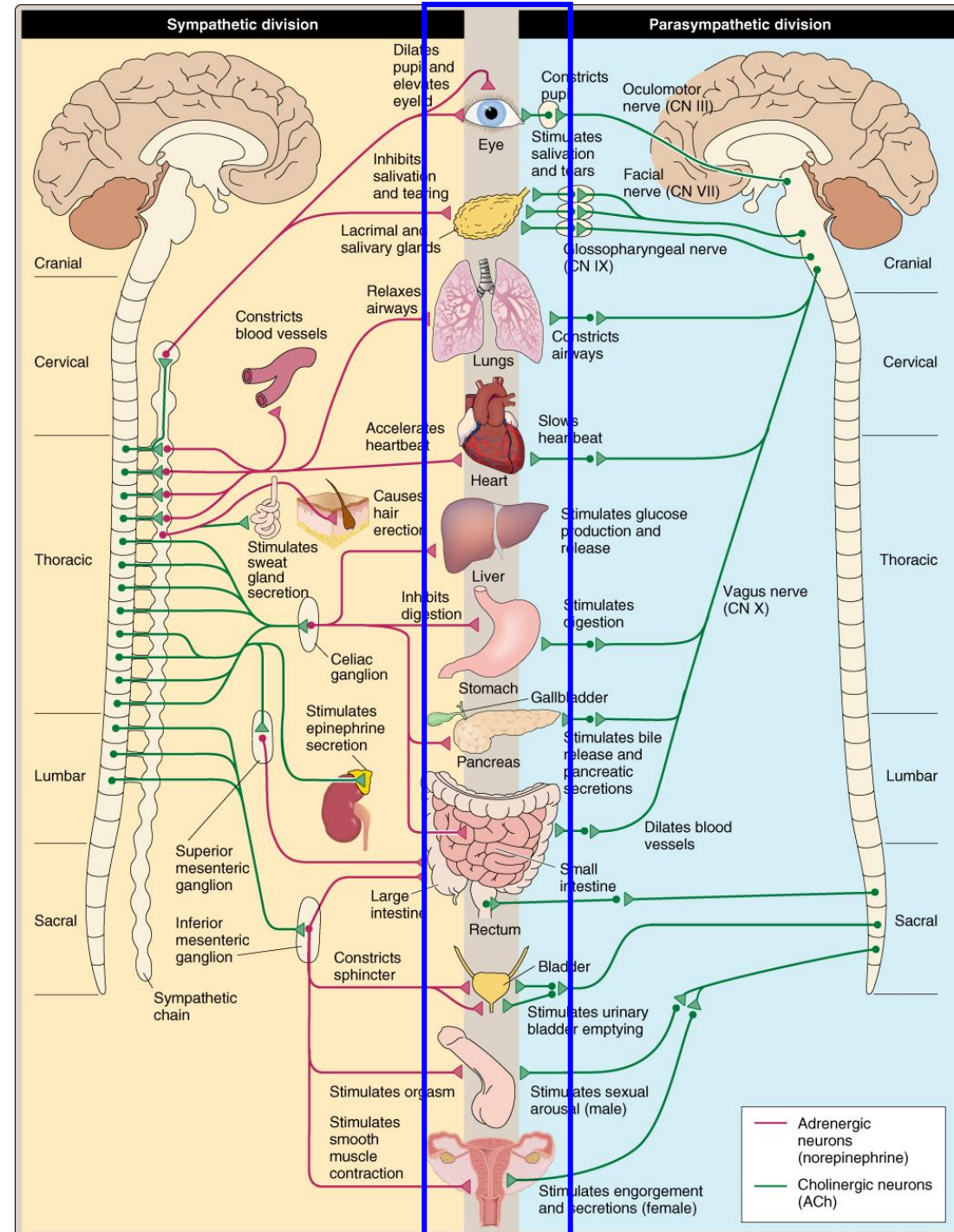
- **SOM.MK.II.BPM1.1.FTM.3.PHYS.0046** Explain the responses of end organs to stimulation of sympathetic and parasympathetic divisions of the autonomic nervous system.
- **SOM.MK.II.BPM1.1.FTM.3.PHYS.0047** List and explain the functions of the eye regulated by the autonomic nervous system, including pupillary constriction and dilation, accommodation, aqueous humor production, and upper eyelid elevation.
- **SOM.MK.II.BPM1.1.FTM.3.PHYS.0048** List the functions, afferent and efferent pathways and stimuli of the baroreflex. Predict clinical implications of baroreflex dysfunction.

LEARNING OBJECTIVES

- **SOM.MK.II.BPM1.1.FTM.3.PHYS.0046** Explain the responses of end organs to stimulation of sympathetic and parasympathetic divisions of the autonomic nervous system.

Dual innervation and antagonistic effects of ANS

There is an equilibrium between sympathetic and parasympathetic effects



List target organs that has **only sympathetic** innervation:

Modified from Bear, M.F., Connors, B.W., and Paradiso, M.A. Neuroscience—Exploring the Brain. Third Edition. Lippincott Williams & Wilkins, 2007

End organ responses to sympathetic and parasympathetic divisions

Region	Effector	Parasympathetic <i>Muscarinic Receptors</i>	Sympathetic <i>Adrenergic Receptors</i>
Skin	Sweat glands	None	Secretion (cholinergic muscarinic receptors)
	Arrector pili muscles	None	Contraction α_1
Blood vessels	Skin (general)	None	Constriction α_1
	Skeletal muscle	None	Dilation β_2
	Viscera	None	Constriction α_1
Heart	SA Node	Decrease heart rate M2	Increase heart rate β_1
	AV Node	Decrease conduction velocity M2	Increase conduction velocity β_1
	Atria	Decrease contractility M2	Increase contractility β_1
	Ventricle	None	Increase contractility β_1

End organ responses to sympathetic and parasympathetic divisions

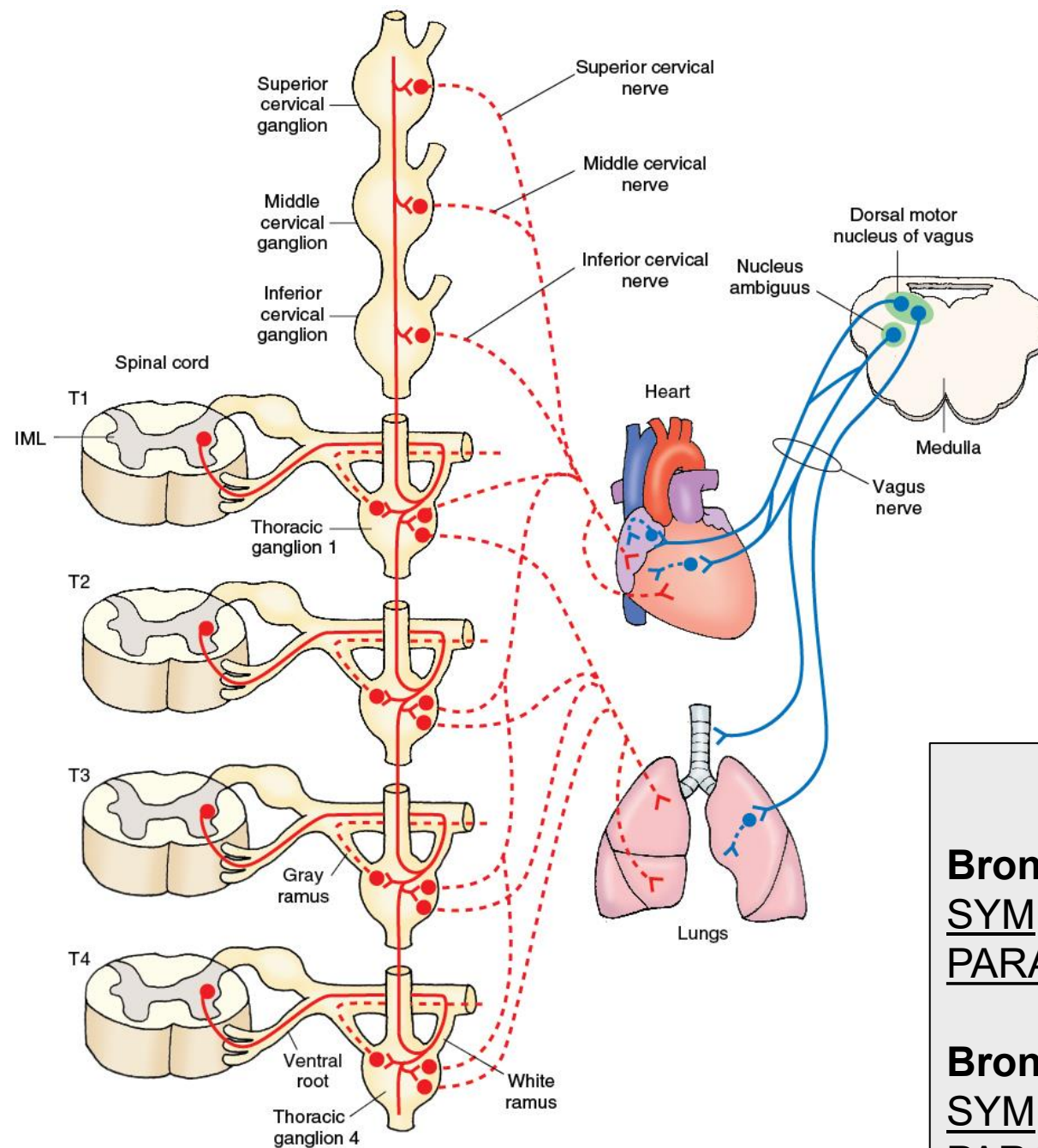
Region	Effector	Parasympathetic <i>Muscarinic Receptors</i>	Sympathetic <i>Adrenergic Receptors</i>
Lungs	Bronchioles	Constriction	Dilation β_2
	Glands	Secretion	Decrease
Gastrointestinal tract	Wall muscles	Contraction M3	Relaxation
	Sphincters	Relaxation	Contraction
	Glands	Stimulation M3	Inhibition
Metabolic functions	Liver	None	Glycogen breakdown and new glucose production
	Pancreas (insulin)	Increased secretion	Decreased secretion α_2
	Adipose Cells	None	Lipolysis β_3
	Kidney	None	Renin release β_1
	Adrenal medulla	None	Secretion of catecholamines (cholinergic nicotinic)

End organ responses to sympathetic and parasympathetic divisions

Region	Effector	Parasympathetic <i>Muscarinic Receptors</i>	Sympathetic <i>Adrenergic Receptors</i>
Urinary system	Detrusor	Contraction M3	Relaxation
	Sphincter	Relaxation	Contraction $\alpha 1$
Reproductive system	Uterus	Variable	Relaxation β_2
	Genitalia	Erection (ACh & NO) M3	Ejaculation
Eye	Circular muscle	Contraction M3	<i>None</i>
	Ciliary muscle	Contraction M3	<i>None</i>
	Radial muscle	None	<i>Contraction $\alpha 1$</i>
	Ciliary epithelium	None	<i>Secretion of aqueous humor $\beta_{1,2}$</i>
	Eyelid muscle (superior tarsal muscle)	None	<i>Elevates upper eyelid</i>

Circular muscle is also known as Sphincter pupillae muscle;
Radial muscle is also known as Dilator pupillae.

HEART & LUNGS



HEART

Sinoatrial node: Heart Rate

SYM: increases β_1

PARASYM: decreases M_2

Atrioventricular node:

Conduction velocity

SYM: increases β_1

PARASYM: decreases M_2

Ventricular muscle

SYM: increases contractility β_1

LUNGS

Bronchial smooth muscle

SYM: bronchodilation β_2

PARASYM: bronchoconstriction M_3

Bronchial glands

SYM: decrease secretion β_2

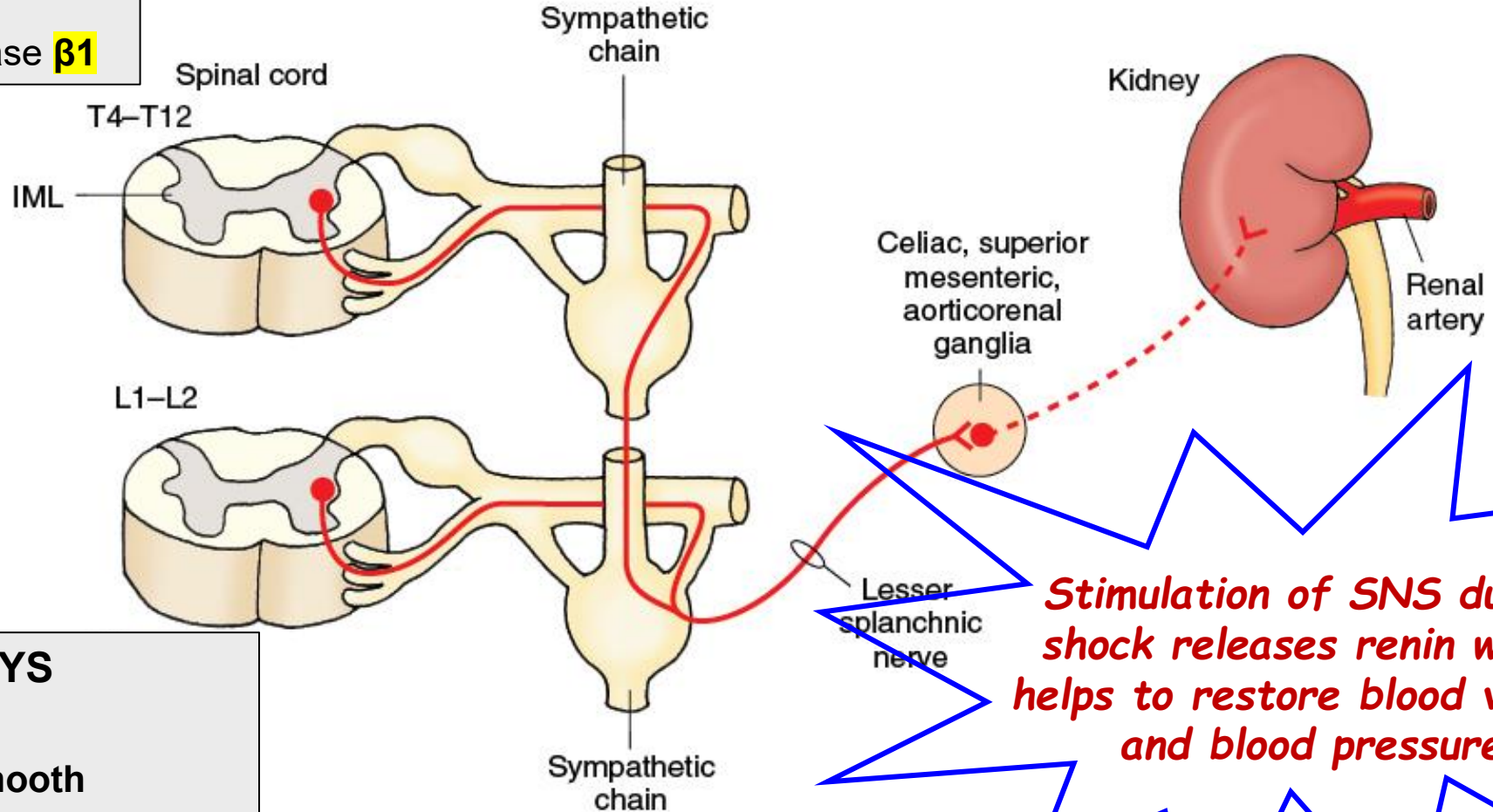
PARASYM: increase secretion M_3

KIDNEYS

KIDNEYS

Juxtaglomerular cells:

SYM: ↑ Renin release **β1**



KIDNEYS

Renal vascular smooth muscles:

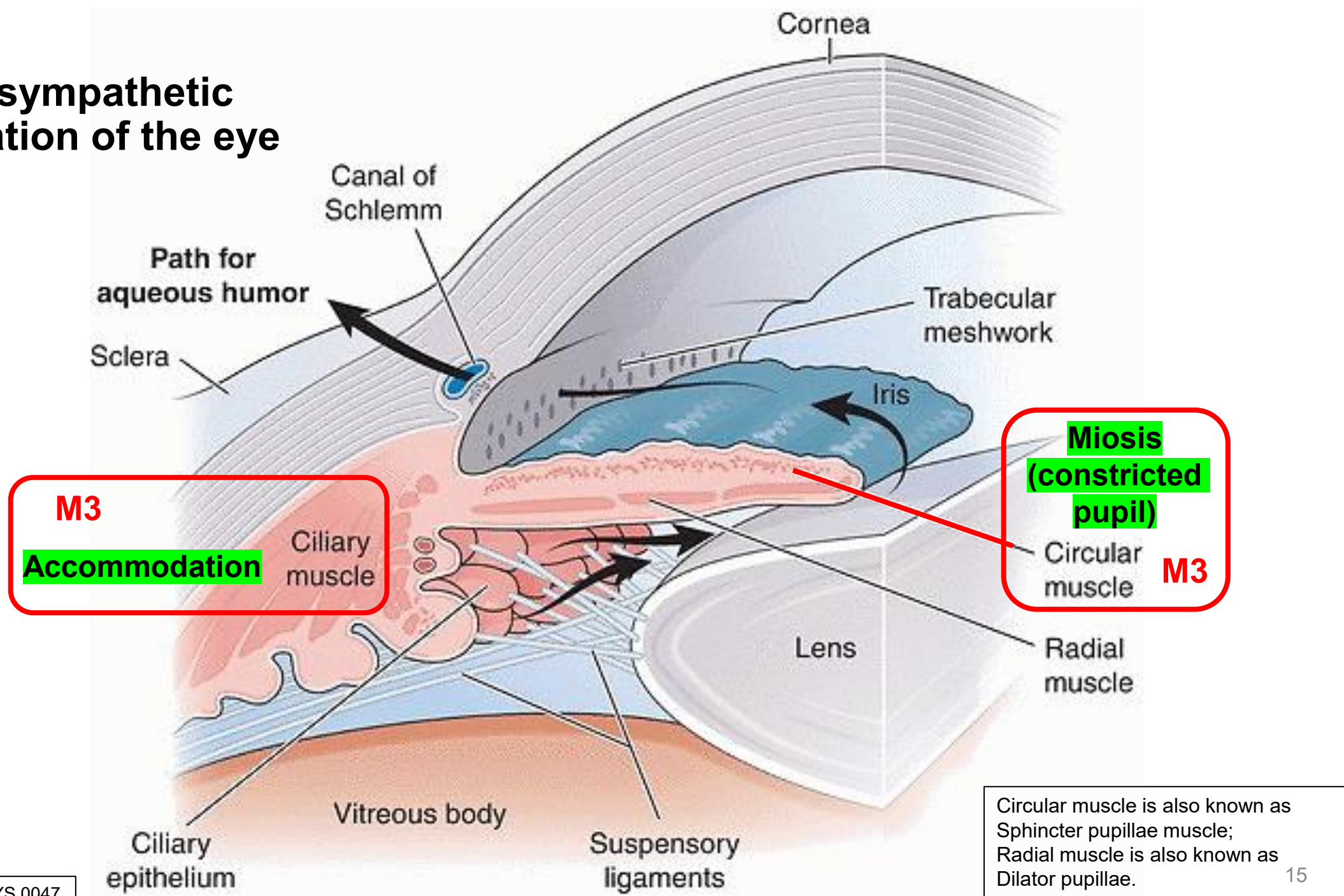
SYM: Relax smooth muscles **D1**

LEARNING OBJECTIVES

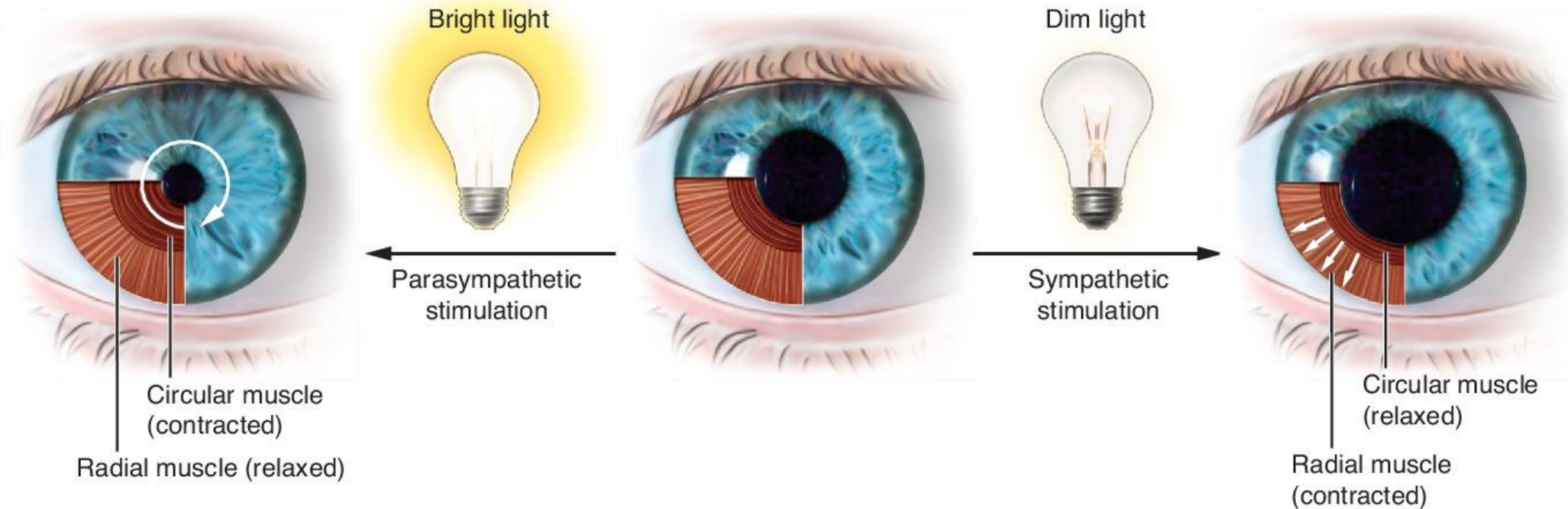
Autonomic Integration

- **SOM.MK.II.BPM1.1.FTM.3.PHYS.0047** List and explain the functions of the eye regulated by the autonomic nervous system, including pupillary constriction and dilation, accommodation, aqueous humor production, and upper eyelid elevation.

Parasympathetic Innervation of the eye



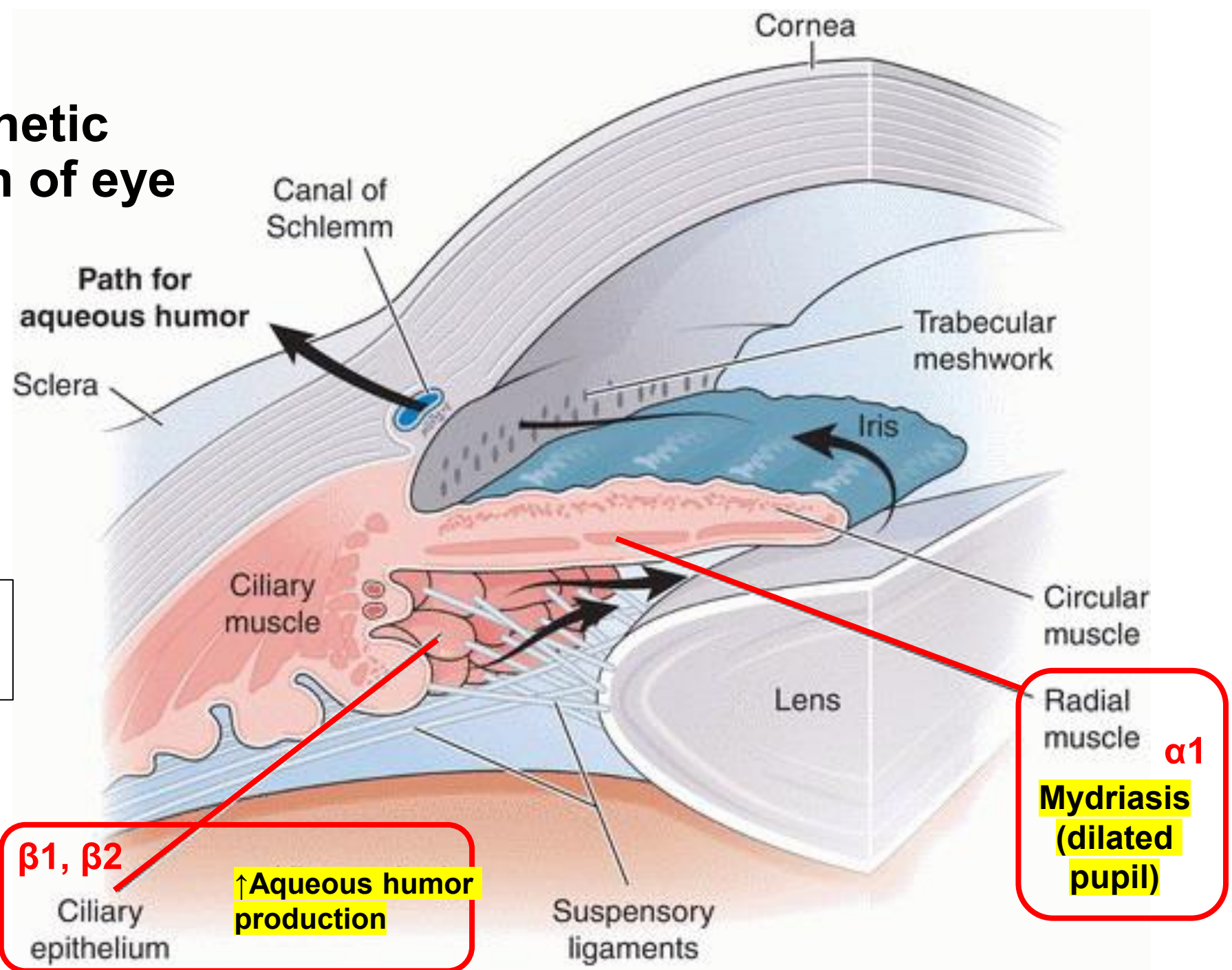
Autonomic regulation of pupil light adaptation



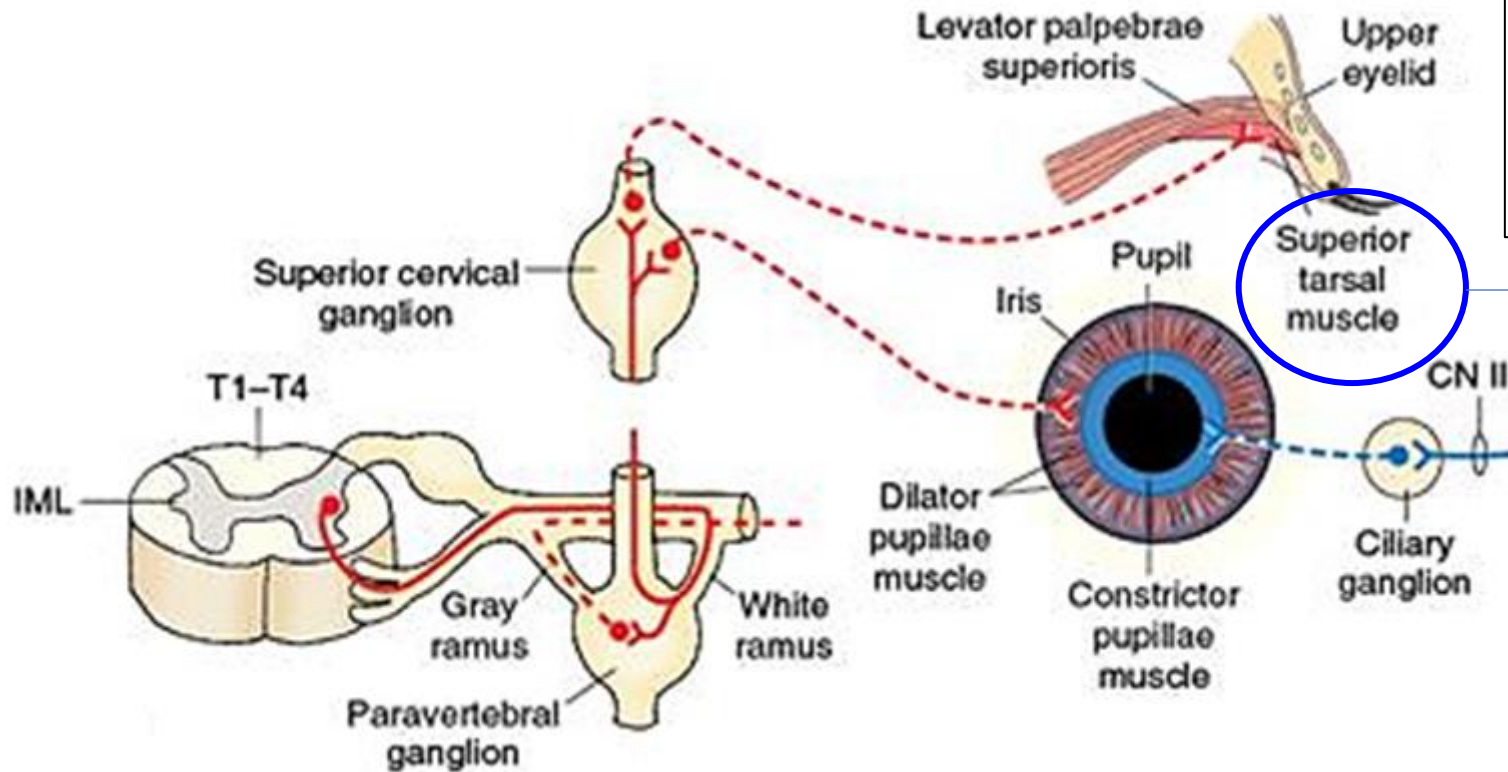
Pupil size adapts to changing light conditions

Circular muscle is also known as Sphincter pupillae muscle;
Radial muscle is also known as Dilator pupillae.

Sympathetic innervation of eye



Autonomic innervation of the face and eye: Supplied by the superior cervical sympathetic nerves



Eyelid elevation is by contraction of 2 muscles:

- Levator palpebrae superioris
- Superior tarsal muscle**-innervated by cervical sympathetic nerves



Patient with hyperthyroidism

Staring look in hyperthyroid patient is partly because of contraction of this muscle!

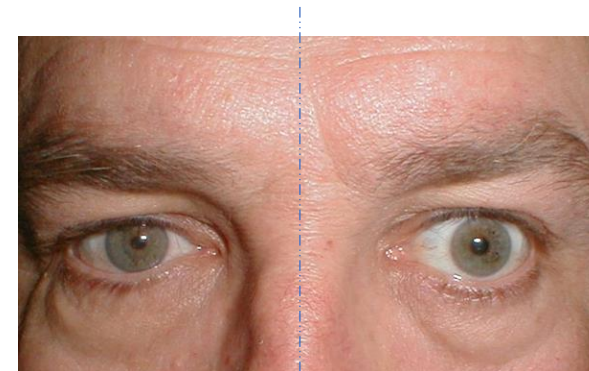
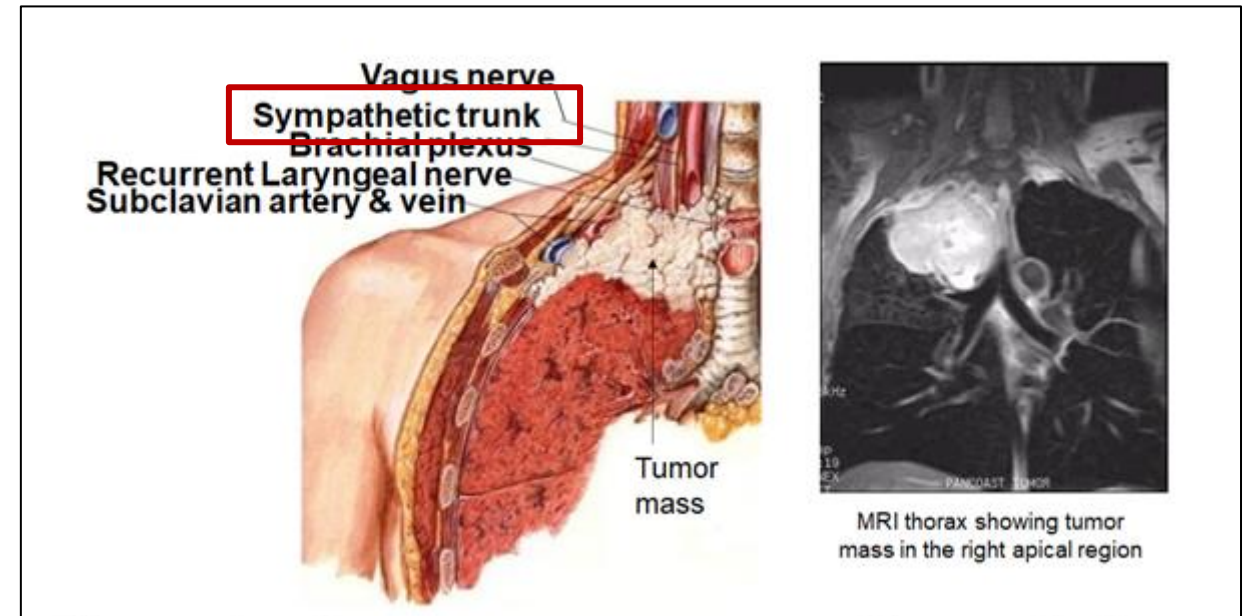
Horner syndrome: Sympathetic denervation of the face because of a tumor in the apical region of the lung

A tumor in the apical region causes damage to the **cervical** sympathetic ganglia

Symptoms:

On the side of the face that is supplied by the damaged nerve

- Ptosis (drooping) of the upper eyelid
- Anhidrosis (absence of sweating) in the face
- Miosis (pupil constriction)



Affected side of face

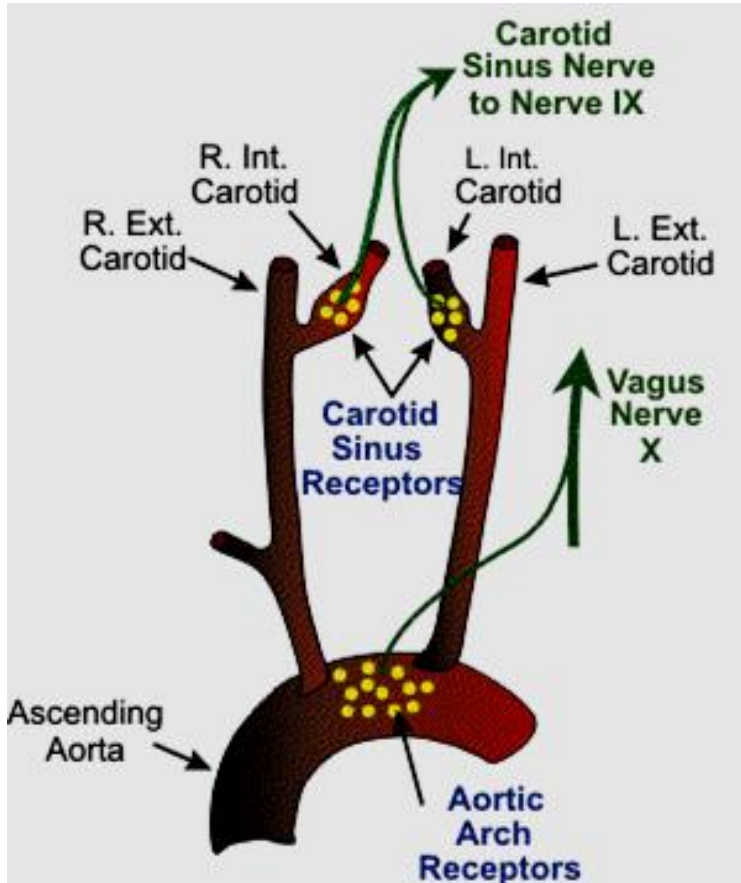
Normal side of face

LEARNING OBJECTIVES

Autonomic Integration

- **SOM.MK.II.BPM1.1.FTM.3.PHYS.0048** List the functions, afferent and efferent pathways and stimuli of the baroreflex. Predict clinical implications of baroreflex dysfunction.

Arterial Baroreceptors initiate Baroreflex to regulate BP



Increase in Blood Pressure (**stimulus**)



Stretching of baroreceptors (**Sensors**)



Increased action potentials in the afferent nerves connected to the baroreceptors (**Afferent n**)

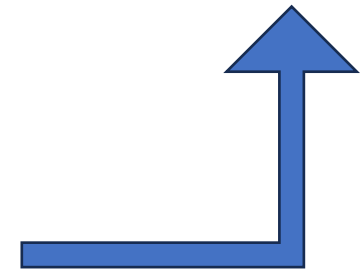


Cardiovascular regulatory centers in brain stem (**Center**)

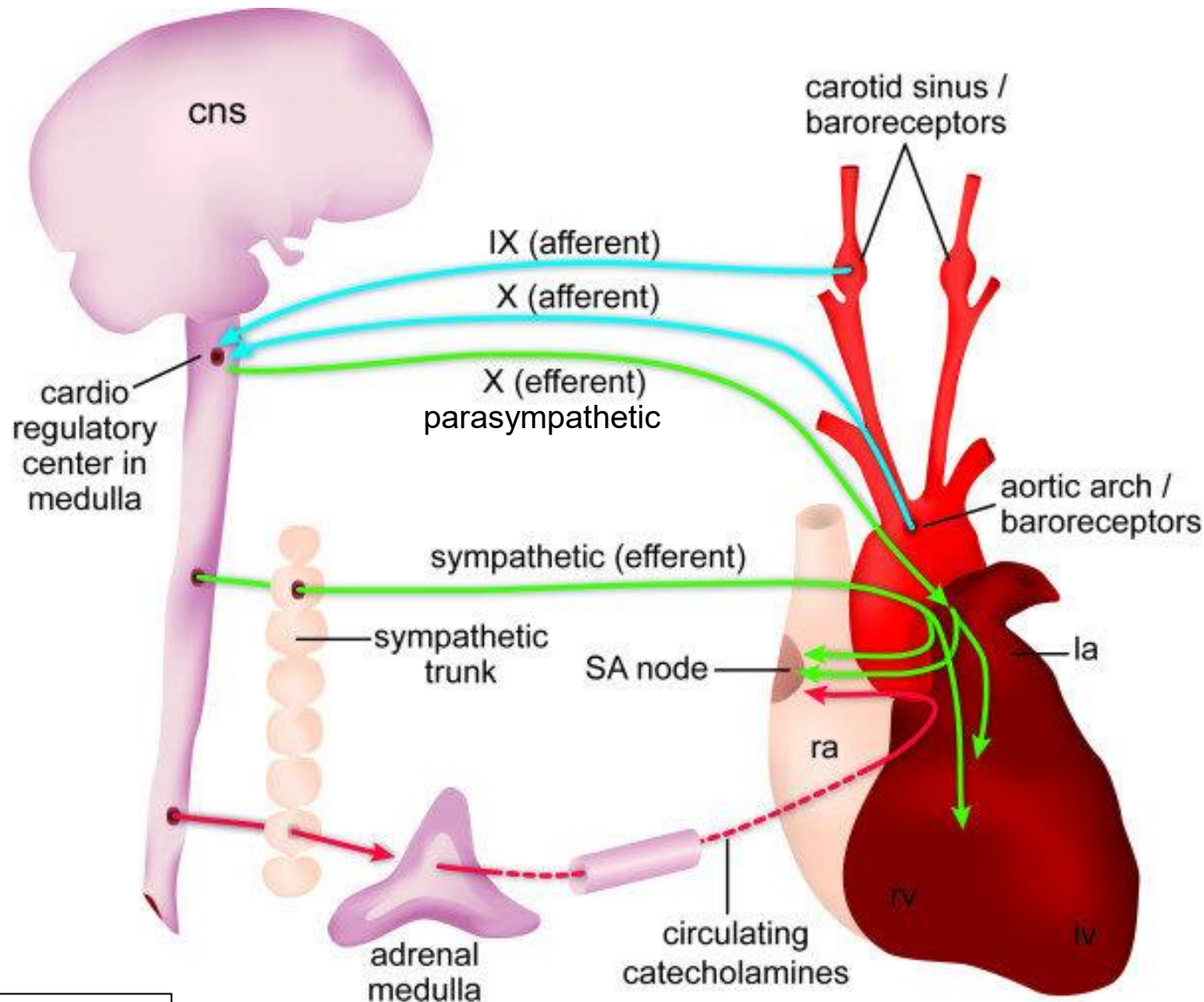


Influence on Autonomic NS (**Efferent n**)

BP back to normal
(**Effect**)



Arterial Baroreceptors regulate blood pressure (BP)

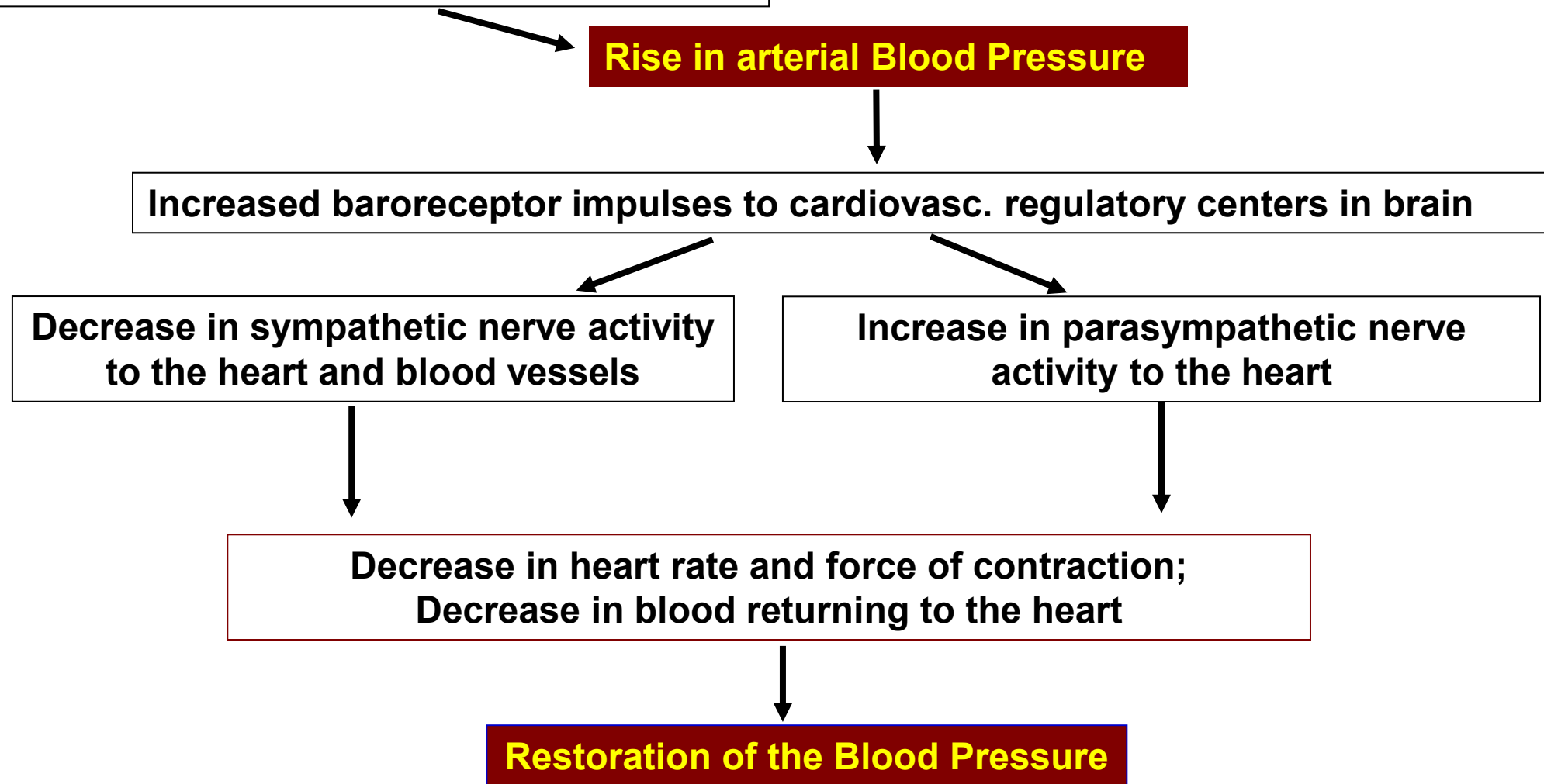


Baroreceptors are sensors located in carotid sinus and aortic arch

Blue: Afferent nerves;
Green: Efferent nerves;
Red: Efferent nerve supplying adrenal medulla

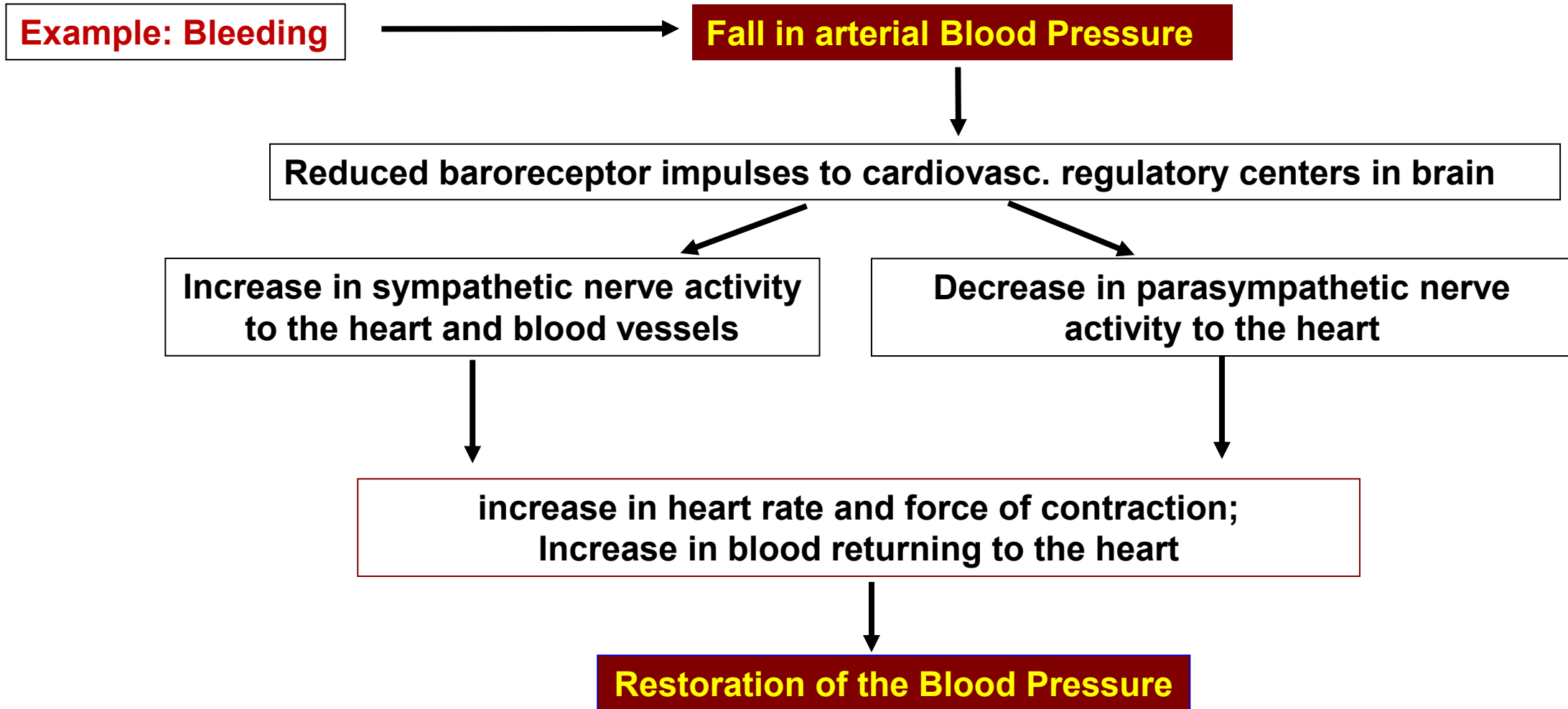
Baroreflex response to a sudden rise in BP

Example: Change posture from standing to supine



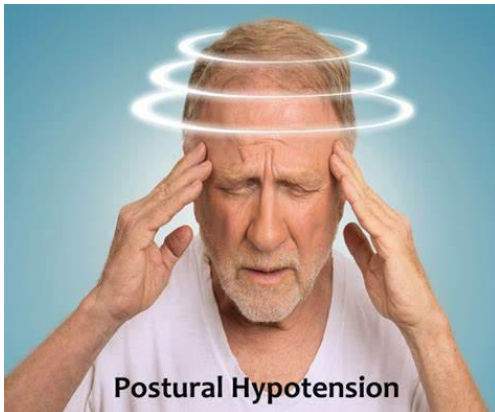
Task: Draw a similar flow chart to show baroreflex response to a **sudden decrease** in the Blood Pressure

Baroreflex response to a sudden lowering BP (Bleeding)



Why baroreflex is important in day-to-day life?

- ❖ Patient with dysfunction of the baroreflex is unable to maintain normal blood pressure.
- ❖ This leads to low blood pressure while changing body postures called **orthostatic hypotension (postural hypotension)**
- ❖ Patient complaints:
 - Lightheadedness
 - Blurry vision
 - Vertigo/dizziness
 - Transient loss of consciousness (syncope)



Other Recommended readings

SIEGEL A, SAPRU HN (2019). Essential neuroscience. Philadelphia, Wolters Kluwer Health/Lippincott Williams & Wilkin

<https://meded.lwwhealthlibrary.com/content.aspx?bookId=2482§ionId=196989456&resultClick=1>

Moore KL, Dalley AF II; Agur AMR. Clinically Oriented Anatomy, 8e
<https://meded.lwwhealthlibrary.com/book.aspx?bookid=2212>

Le, Tao and Bhushan, Vikas. First Aid for the USMLE Step 1 2025, Thirty Fifth edition. New York: McGraw-Hill Education, 2025.

**GET READY FOR CLICKER-BASED INTERACTIVE
FLIPPED CLASSROOM SESSION**